



INFORMATION TECHNOLOGY INSTITUTE

ANALOG IC DESIGN

Lab 1

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1. Part 2: Series RLC Circuit

1.1. A) Input Impedance Expression

For R , L , C in series, the input impedance is the sum of the three branch impedances:

$$Z_{IN}(s) = R + sL + \frac{1}{sC} \quad (1)$$

$$Z_{IN}(s) = \frac{s^2LC + sRC + 1}{sC} \quad (2)$$

$$\boxed{Z_{IN}(s) = \frac{L}{sC} \left(s^2 + \frac{R}{L}s + \frac{1}{LC} \right)} \quad (3)$$

Comparing the bracketed numerator to the canonical form $s^2 + \frac{\omega_0}{Q}s + \omega_0^2$:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} \quad (4)$$

At $s = j\omega_0$, $j\omega_0 L$ and $1/(j\omega_0 C)$ cancel exactly, so $Z_{IN}(j\omega_0) = R$ — the *minimum* magnitude of Z_{IN} , with zero phase.

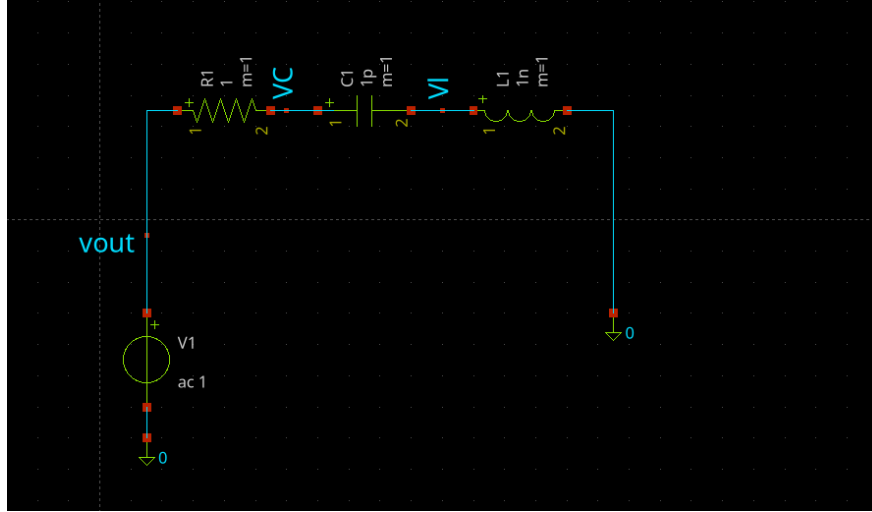


Figure 1: Series RLC circuit: R_1 , C_1 , L_1 in series, driven by an AC voltage source V_1 , with V_C and V_L probed across the capacitor and inductor.

The circuit was simulated with $R = 1 \Omega$, $L = 1 \text{ nH}$, $C = 1 \text{ pF}$, giving

$$Q = \frac{1}{1} \sqrt{\frac{10^{-9}}{10^{-12}}} = \sqrt{1000} \approx 31.6 \quad (5)$$

1.2. B) AC Analysis: $|Z_{IN}|$ vs. Frequency

With $R = 1\ \Omega \ll \omega_0 L \approx 31.6\ \Omega$ over most of the band, $|Z_{IN}| \approx |X_L - X_C|$ except very near resonance, producing a sharp, clearly visible V-shaped dip.

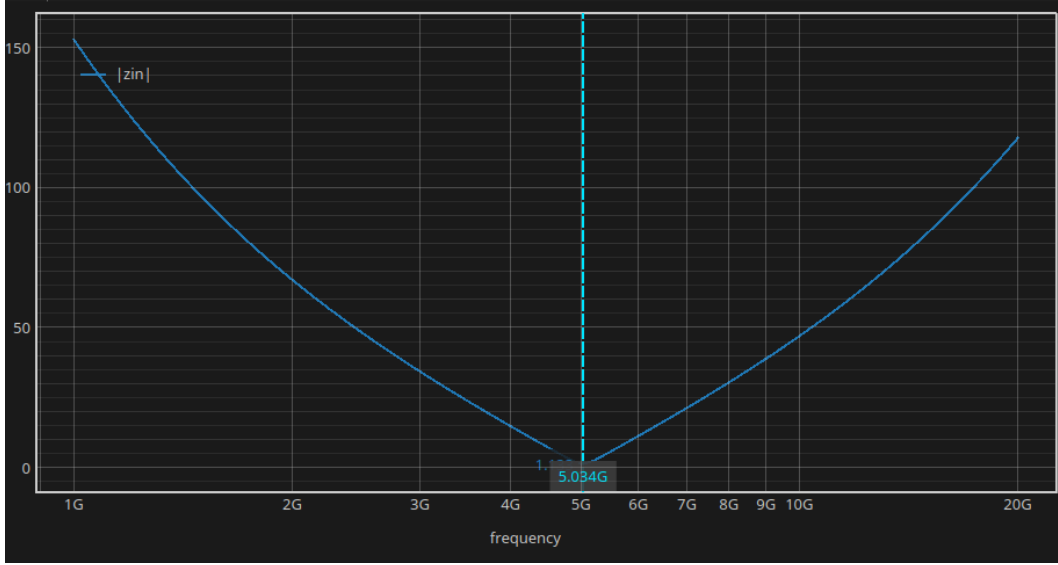


Figure 2: $|Z_{IN}|$ vs. frequency ($R = 1\ \Omega$): $\approx 153\ \Omega$ at 1 GHz, dipping to $\approx 1\ \Omega$ at $f_0 \approx 5.03\ \text{GHz}$, rising to $\approx 118\ \Omega$ at 20 GHz.

Inductive vs. Capacitive Regions

Below resonance ($f < 5.03\ \text{GHz}$), $X_C = 1/(\omega C)$ exceeds $X_L = \omega L$, so the net reactance is negative and the impedance is capacitive – $|Z_{IN}|$ falls as frequency increases toward f_0 . Above resonance, X_L dominates, the impedance is inductive, and $|Z_{IN}|$ rises with frequency. This is the opposite labeling convention from the parallel case in Part 1, since here it is the *series reactance* that changes sign, not the dominant admittance branch.

1.3. C) Resonance Frequency Verification

Analytically,

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \approx 5.033\ \text{GHz} \quad (6)$$

The simulated minimum in $|Z_{IN}|$ occurs at $F0 = 5.034\ \text{GHz}$ (Fig. 3).

1.4. D) Quality Factor Evaluation

Analytically,

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \sqrt{\frac{10^{-9}}{10^{-12}}} \approx 31.6 \quad (7)$$

```
Using SPARSE 1.3 as Direct Linear Solver
Note: v1: has no value, DC 0 assumed

No. of Data Rows : 301
f0 = 5.011872e+09 with= 1.034533e+00
f1 = 4.972897e+09
fh = 5.067100e+09
f0 = 5.011872e+09
f1 = 4.972897e+09
fh = 5.067100e+09
bw_sim = 9.420300e+07
q_sim = 5.320289e+01
binary raw file "rlc_02.raw"
ngspice 1 -> █
```

Figure 3: Simulator-extracted $F0 = 5.0119$ GHz, $F1 = 4.9729$ GHz, $Fh = 5.0671$ GHz, $BW_{sim} = 94.2$ MHz, $Q_{sim} = 53.2$.

The simulated $Q_{sim} = 53.2$ is substantially higher than the analytical $Q = 31.6$. The reason is sampling resolution: with 60 points/decade over the swept 1–20 GHz range.

```
No. of Data Rows : 1302
f0 = 5.035194e+09 with= 1.000407e+00
f1 = 4.953996e+09
fh = 5.113032e+09
f0 = 5.035194e+09
f1 = 4.953996e+09
fh = 5.113032e+09
bw_sim = 1.590360e+08
q_sim = 3.166072e+01
binary raw file "rlc_02.raw"
ngspice 1 -> █
```

Figure 4: Simulator-extracted with 1000 point per decade $Q = 31.6$

1.5. E) Bandwidth

Analytically,

$$BW = \frac{R}{2\pi L} = \frac{1}{2\pi(10^{-9})} \approx 159.15 \text{ MHz} \quad (8)$$

equivalently $BW = f_0/Q = 5.033 \text{ GHz}/31.6 \approx 159.2 \text{ MHz}$.

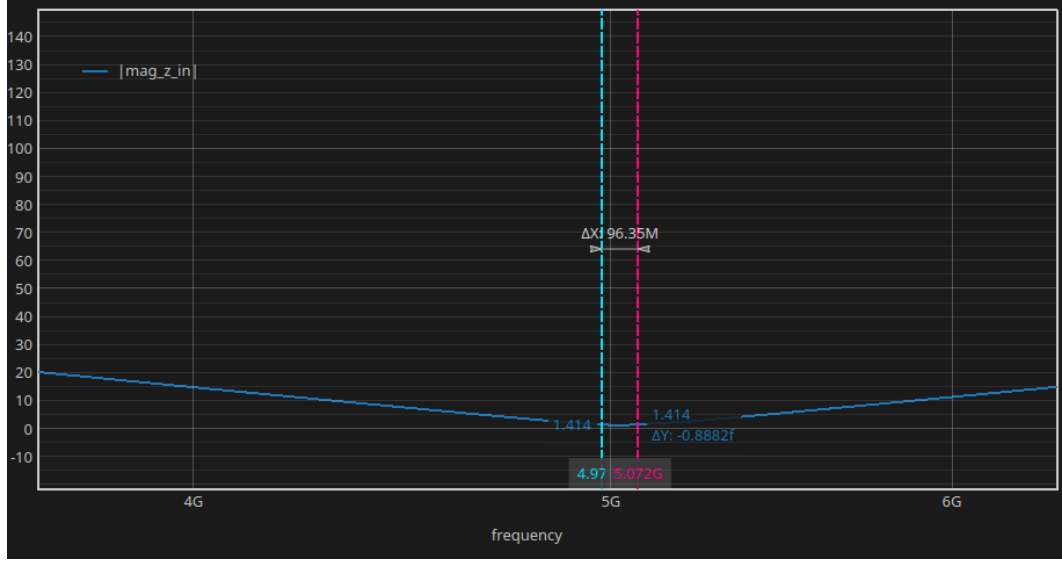


Figure 5: Zoomed view near resonance: -3 dB points (where $|Z_{IN}|$ rises to $R\sqrt{2} \approx 1.414 \Omega$) at 4.97 GHz and 5.072 GHz, giving $BW_{sim} = \Delta X \approx 94$ –96 MHz.

The simulated bandwidth (≈ 94.2 MHz) is roughly 41% narrower than the analytical prediction (159.15 MHz), for the same under-sampling reason discussed in Part D: at this Q , the -3 dB bandwidth is narrower than the frequency step between adjacent simulation points, so the true bandwidth cannot be resolved precisely by this sweep.

1.6. F) Parametric Sweep on R

R was swept over $\{1, 10, 100, 1000\} \Omega$ (a decade sweep, to capture the full range from very underdamped to heavily overdamped behavior) across the same 1–20 GHz band.

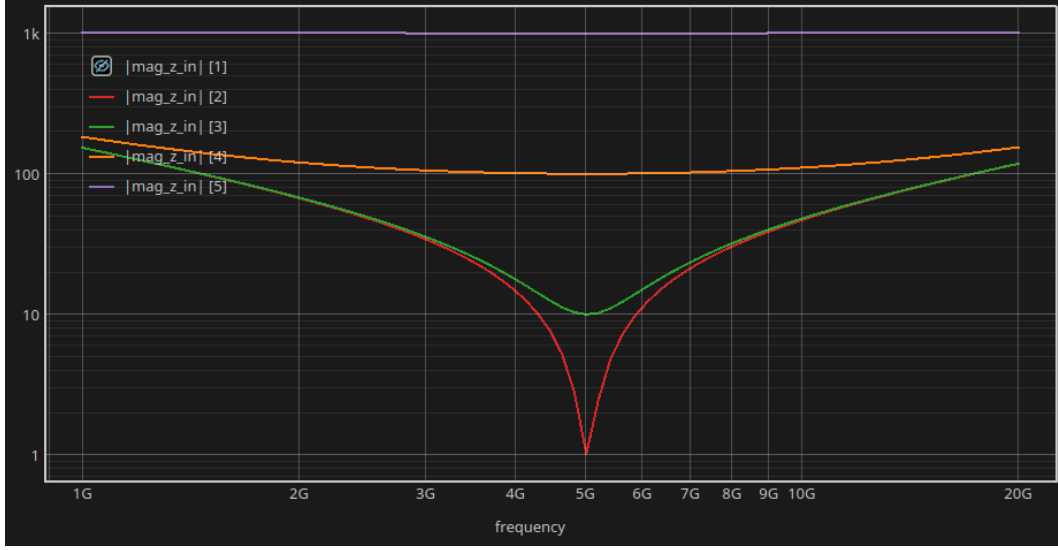


Figure 6: $|Z_{IN}|$ overlaid for $R = 1, 10, 100, 1000 \Omega$ (log scale). All curves share the same $f_0 \approx 5.03$ GHz dip location; dip depth and sharpness vary strongly with R .

Effect of Varying Resistance (R)

Since $\omega_0 = 1/\sqrt{LC}$ is independent of R , all curves dip at the same f_0 . Because $Q = \frac{1}{R}\sqrt{L/C}$ is *inversely* proportional to R – the four decades of R give four very different Q values:

$R (\Omega)$	Q
1	31.6
10	3.16
100	0.316
1000	0.0316

The $R = 1 \Omega$ curve dips sharply almost all the way to 1Ω at f_0 (very high Q); the $R = 10 \Omega$ curve still shows a clear but shallower dip; the $R = 100 \Omega$ curve is nearly flat with only a slight dip near f_0 ; and the $R = 1 \text{ k}\Omega$ curve is essentially flat across the whole band, since at $Q = 0.0316$ the reactive elements barely perturb the dominant resistive impedance. This confirms that lower R (higher Q) produces a sharper, deeper resonance dip in the series topology, exactly the opposite trend to Part 1's parallel circuit, where higher R gave the sharper peak.

1.7. G) Transient Response at Resonance

A sine-wave voltage source of amplitude 1 V at $f_0 = 5.03$ GHz was applied with $R = 1 \Omega$, and V_L , V_C were plotted.

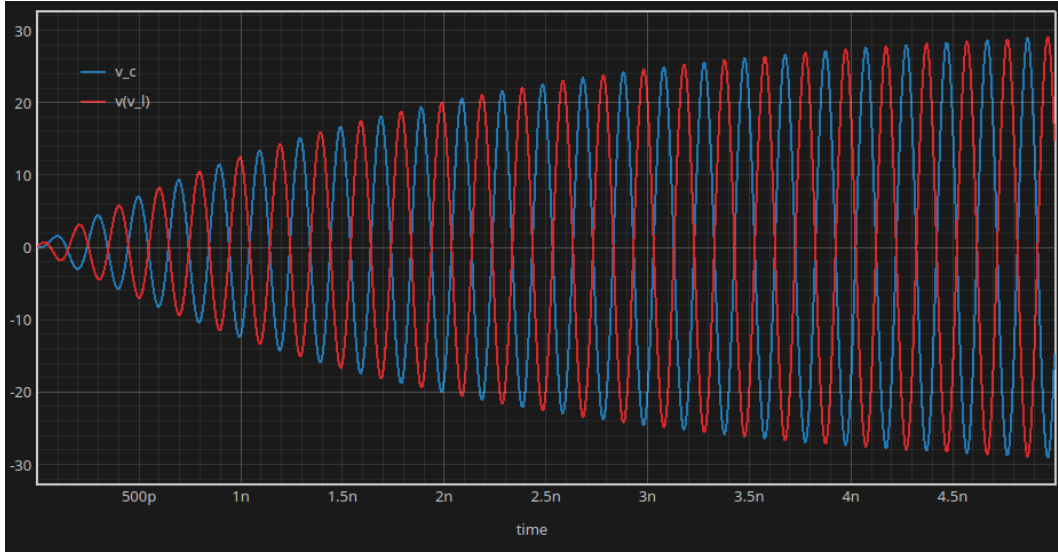


Figure 7: Transient response at resonance, $R = 1\ \Omega$: V_L and V_C build up to $\approx 28\text{--}30\text{ V}$, a passive voltage gain of $\approx Q$.

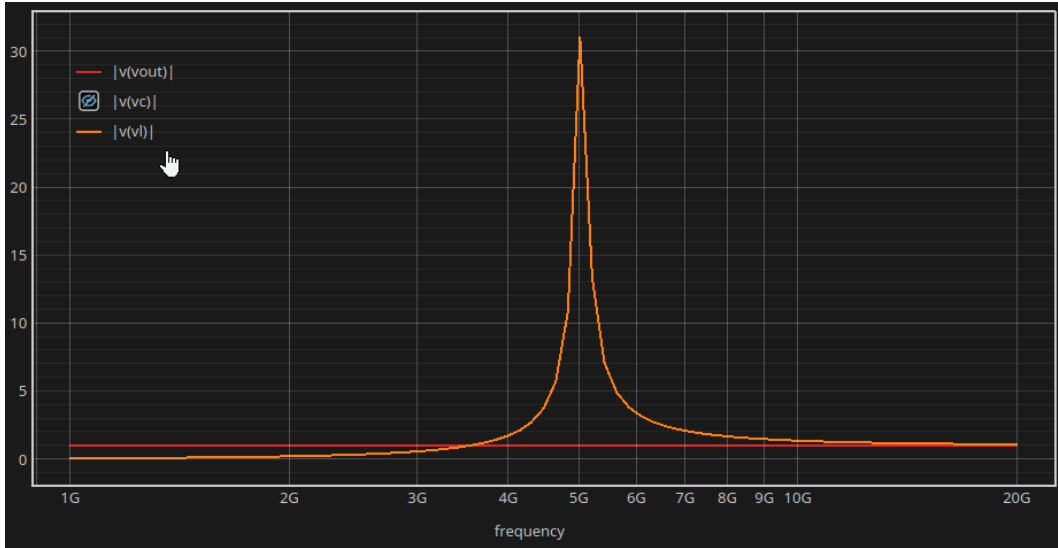


Figure 8: AC sweep of V_L/V_{in} ($R = 1\ \Omega$): gain peaks at ≈ 31 near f_0 , confirming the time-domain amplification.

Transient & AC Gain Analysis

Amplitude: at resonance, $I = V_0/R$ flows through all three elements, so

$$V_{L,C} = I \times \omega_0 L = \frac{V_0}{R} \omega_0 L = V_0 \times Q \approx 1 \text{ V} \times 31.6 \approx 31.6 \text{ V}$$

matching both the transient buildup in Fig. 7 and the AC peak gain in Fig. 8. A passive circuit cannot amplify power, but at resonance the same current is forced through a much larger reactive impedance than the source impedance, so the *voltage* across L or C individually exceeds the source voltage by a factor of Q .

Phase relationship: V_L and V_C are 180° out of phase at exact resonance, since $j\omega_0 L = -1/(j\omega_0 C)$ – they cancel in the loop, leaving the source voltage entirely across R .

Capacitor voltage: $V_{C,ss} = I \times \frac{1}{\omega_0 C} = Q \times V_0 \approx 31.6 \text{ V}$, reached gradually with a buildup time constant $\tau \approx 2Q/\omega_0 \approx 2 \text{ ns}$, matching the growing envelope in Fig. 7.

1.8. H) Step Response with Parametric Sweep on R

A step voltage source was applied while sweeping R over $\{1, 10, 100, 1000\} \Omega$; V_C is plotted.

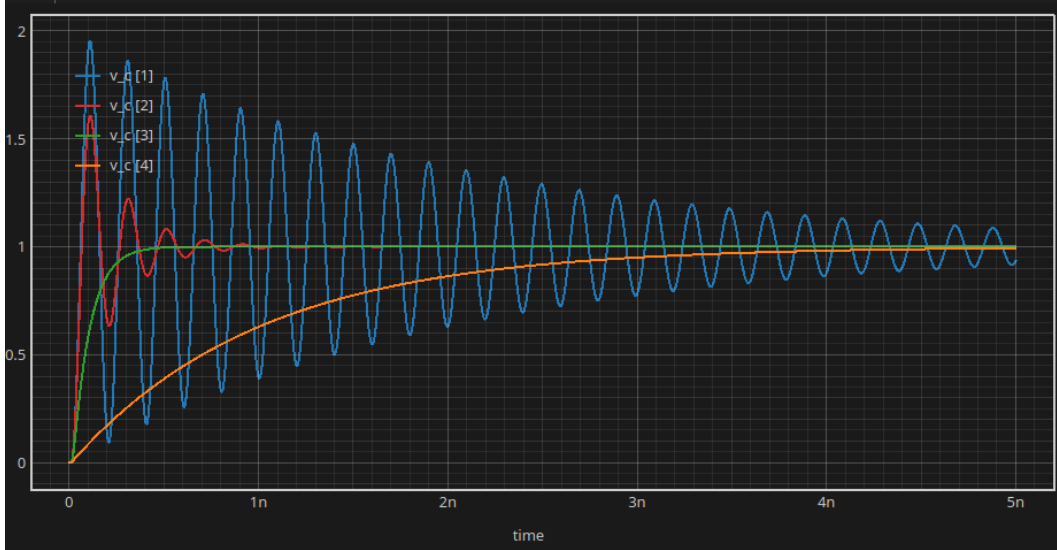


Figure 9: Step response of V_C for $R = 1, 10, 100, 1000 \Omega$.

Damping Behavior Observations

The damping factor is $\alpha = R/(2L)$. The four traces span the full range of damping behavior:

- $R = 1\ \Omega$ ($Q = 31.6$): strongly underdamped, ringing continuously with a large overshoot to $\approx 2\times$ the final value, decaying very slowly.
- $R = 10\ \Omega$ ($Q = 3.16$): still underdamped, with a couple of visible oscillations before settling within $\sim 1\ \text{ns}$.
- $R = 100\ \Omega$ ($Q = 0.316$): close to critically damped, rising smoothly to steady state with little to no overshoot.
- $R = 1\ \text{k}\Omega$ ($Q = 0.0316$): heavily overdamped, rising monotonically but noticeably *slower* than the $R = 100\ \Omega$ case.

This last point highlights a subtlety: although larger R gives larger α (naively "more damping"), for an overdamped ($\alpha \gg \omega_0$) system the two poles split into a fast pole near -2α and a slow, dominant pole near $-\omega_0^2/(2\alpha) \approx -1/(RC)$. As R increases, this dominant pole moves *closer* to the origin (time constant $\sim RC$ grows), so the response settles *more slowly* for larger R , exactly as seen comparing the $R = 100\ \Omega$ and $R = 1\ \text{k}\Omega$ traces in Fig. 9. This mirrors the same $Q \propto 1/R$ trend already observed in the frequency domain (Part F).

- If $Q < 0.5$ ($\zeta > 1$): overdamped system, roots are real, negative, and distinct, like two first-order RC filters in cascade
- If $Q = 0.5$ ($\zeta = 1$): critical damped system, roots are real, negative, and equal
- If $Q > 0.5$ ($\zeta < 1$): underdamped system, roots are complex conjugate

Figure 10: Damping behavior across swept resistance ranges.

2. [LAB 1] g_m and r_o in Triode and Saturation

2.0.1 Hand Analysis

The output resistance r_o is defined as:

$$r_o = \left(\frac{\partial I_D}{\partial V_{DS}} \right)^{-1} \quad (9)$$

In the **triode region**, I_D is strongly dependent on V_{DS} :

$$I_D \approx \mu_n C_{ox} \frac{W}{L} \left((V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right) \quad (10)$$

Taking the derivative for transconductance gives $g_m \approx \mu_n C_{ox} \frac{W}{L} V_{DS}$, meaning g_m increases linearly with V_{DS} while r_o remains very small.

In the **saturation region**, the channel pinches off and I_D is ideally independent of V_{DS} (ignoring CLM):

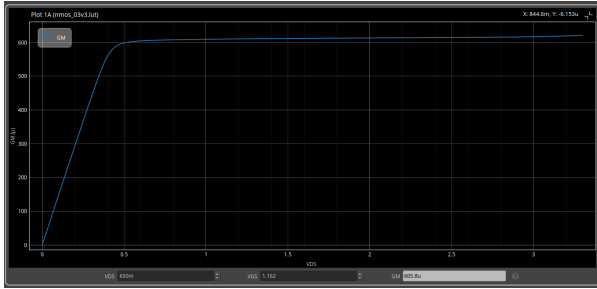
$$r_o \approx \frac{1}{\lambda I_D} \quad (11)$$

Here, g_m saturates to a constant value dependent on V_{GS} , and r_o becomes large, ideally increasing linearly with V_{DS} as CLM predicts.

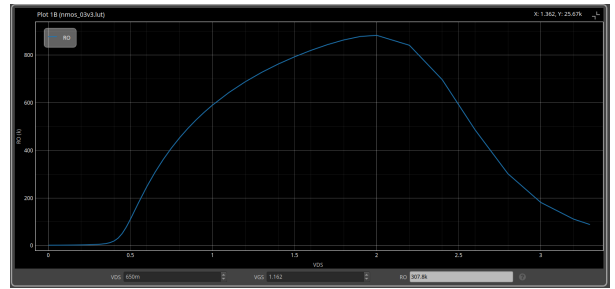
2.0.2 Simulation Setup

The long-channel device ($L = 2.8 \mu\text{m}$, $W/L = 10$) was biased at $V_{GS} = 1.162 \text{ V}$, chosen to satisfy $V_{GS} \approx V_{TH} + 0.5 \text{ V}$ (estimated from the I_D - V_{GS} sweep), and V_{DS} was swept from 0 to $V_{DD} = 3.3 \text{ V}$.

2.0.3 Simulation Results



(a)



(b) Output Resistance (r_o) vs V_{DS}

Figure 11: Small-signal parameters behavior across operating regions ($V_{GS} = 1.162 \text{ V}$, $L = 2.8 \mu\text{m}$).

At $V_{DS} = 0.65 \text{ V}$, the simulator reports $g_m \approx 605.8 \mu\text{A/V}$ and $r_o \approx 307.8 \text{ k}\Omega$, consistent with the device being just past the triode-saturation boundary. r_o continues to rise with V_{DS} , reaching a maximum of roughly $900 \text{ k}\Omega$ near $V_{DS} \approx 1.9\text{--}2.0 \text{ V}$, before decreasing at higher V_{DS} .

1) Variation of g_m vs V_{DS} :

- **In the first part of the curve, is the relation linear? Why?** Yes. In the deep triode region, the drain current is approximately $I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) V_{DS}$. Since transconductance is defined as $g_m = \frac{\partial I_D}{\partial V_{GS}}$, taking the derivative yields $g_m \approx \mu_n C_{ox} \frac{W}{L} V_{DS}$. This shows that g_m is directly proportional to V_{DS} , resulting in a linear relationship.
- **Does g_m saturate? Why?** Yes. Once V_{DS} exceeds the overdrive voltage (around 0.4–0.5 V here) and the device enters saturation, the channel pinches off and I_D becomes nearly independent of V_{DS} . From the plot, g_m flattens out at about $600 \mu\text{A/V}$ and only creeps up very slightly afterward (a second-order effect from channel-length modulation).
- **Where do you want to operate for analog amplifier applications? Why?** In saturation, past the knee where g_m has flattened. This gives a stable, predictable transconductance that does not vary with the output voltage swing, which is essential for linear, low-distortion voltage amplification.

2) Variation of r_o vs V_{DS} :

- **Does r_o saturate just after the transistor enters saturation, similar to g_m ?** No. Unlike g_m , r_o does not flatten immediately at the triode-saturation boundary. Instead, it keeps rising well into saturation, peaking around $V_{DS} \approx 1.9\text{--}2.0 \text{ V}$ (close to $900 \text{ k}\Omega$), since the depletion region at the drain is still evolving and I_D has not yet become perfectly flat.
- **Does r_o increase if the transistor is biased more into saturation?** Only up to a point. In this simulation r_o increases with V_{DS} up to about 2 V, but then decreases at higher V_{DS} . This roll-off is because V_{DS} is approaching V_{DD} with a fixed V_{GS} , so the device is being pushed toward the edge of its valid operating range (and I_D starts increasing more steeply again), which drags r_o back down.
- **Should we operate the transistor at the edge of saturation?** No. Right at the triode-saturation boundary, r_o is still small and changing rapidly with V_{DS} , so the intrinsic gain is poor and the operating point is sensitive to small variations in bias.
- **Where do you want to operate for analog amplifier applications? Why?** In the middle of the saturation region, comfortably past the knee but before r_o starts rolling off near the supply rail. The intrinsic gain is $A_v = g_m r_o$, so this region — where both g_m is flat and r_o is near its peak (around $V_{DS} \approx 1.5\text{--}2 \text{ V}$ in this case) — maximizes voltage gain while leaving headroom for output swing.

3. [LAB 2] Gain Linearization (Negative Feedback)

3.1. Active Load and Feedback Network Design

To stabilize the open-loop gain and minimize non-linearity due to input signal variations, the passive resistive load is replaced with a PMOS current source (active load). A negative feedback network is introduced utilizing two resistors, R_1 and R_2 .

The ideal closed-loop gain is defined analytically as:

$$A_{CL} \approx -\frac{R_2}{R_1} \ll A_{OL} \approx g_m(r_{op} || r_{on} || R_2) \quad (12)$$

```
Info: [Sizing Assistant] The resultant point 'W = 9.76u'
Info: [Sizing Assistant] The resultant point 'VGS = 818.1m'
Info: [Sizing Assistant] The resultant point 'VTH = 668.4m'
Info: [Sizing Assistant] The resultant point 'GM = 99.17u'
Info: [Sizing Assistant] The resultant point 'GDS = 176.7n'
```

Figure 12: NMOS Sizing Values

```
- Info: [Sizing Assistant] The resultant point 'W = 42.84u'
- Info: [Sizing Assistant] The resultant point 'VGS = 936.8m'
- Info: [Sizing Assistant] The resultant point 'VTH = 774.9m'
- Info: [Sizing Assistant] The resultant point 'GM = 99.03u'
- Info: [Sizing Assistant] The resultant point 'GDS = 112.9n'
```

Figure 13: PMOS Sizing Values

Figure 14: Sizing Assistant results for NMOS and PMOS transistors.

Parameter	NMOS	PMOS
W	9.76 μ	42.84 μ
VGS	818.1 m	936.8 m
VTH	668.4 m	774.9 m
GM	99.17 μ	99.03 μ
GDS	176.7 n	112.9 n

Table 1: Comparison of NMOS and PMOS Sizing Values

Based on the required specification of $|A_v| = 10$, the feedback resistor R_2 is chosen as 10 M Ω to prevent loading the amplifier's intrinsic output resistance, and the input resistor R_1 is set to 1 M Ω .

$$|A_{v, \text{closed}}| \approx \frac{R_f}{R_{in}} \implies R_f = |A_v| \times R_{in} = 10 \times 1 \text{ M}\Omega = 10 \text{ M}\Omega \quad (13)$$

With $R_{in} = 1 \text{ M}\Omega$, choose $R_f = 10 \text{ M}\Omega$.

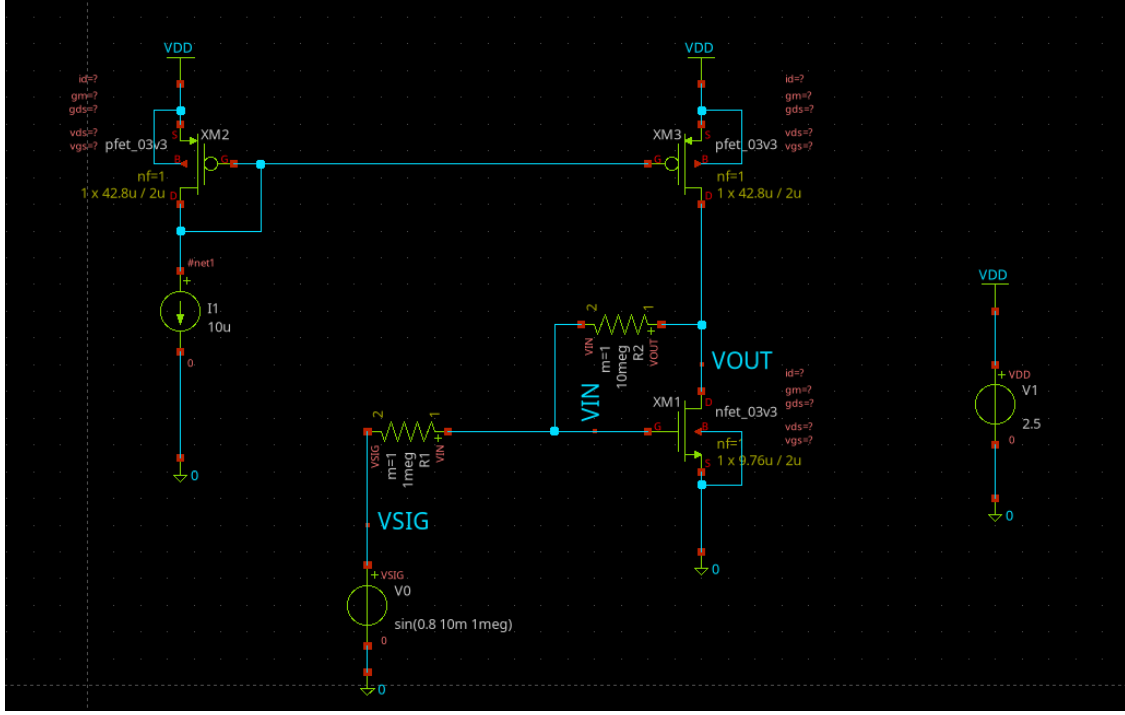


Figure 15: Schematic of the linearized CS Amplifier. M1 (NMOS) is the main amplifier ($1 \times 9.76\mu/2\mu$), and M3 (PMOS) acts as the active load ($1 \times 42.8\mu/2\mu$) biased by the replica branch M2.

3.2. Large Signal DC Transfer Characteristic (VTC)

A DC sweep was performed on the input signal V_{SIG} from 0V to 2.5V to evaluate the large-signal behavior of the linearized amplifier.

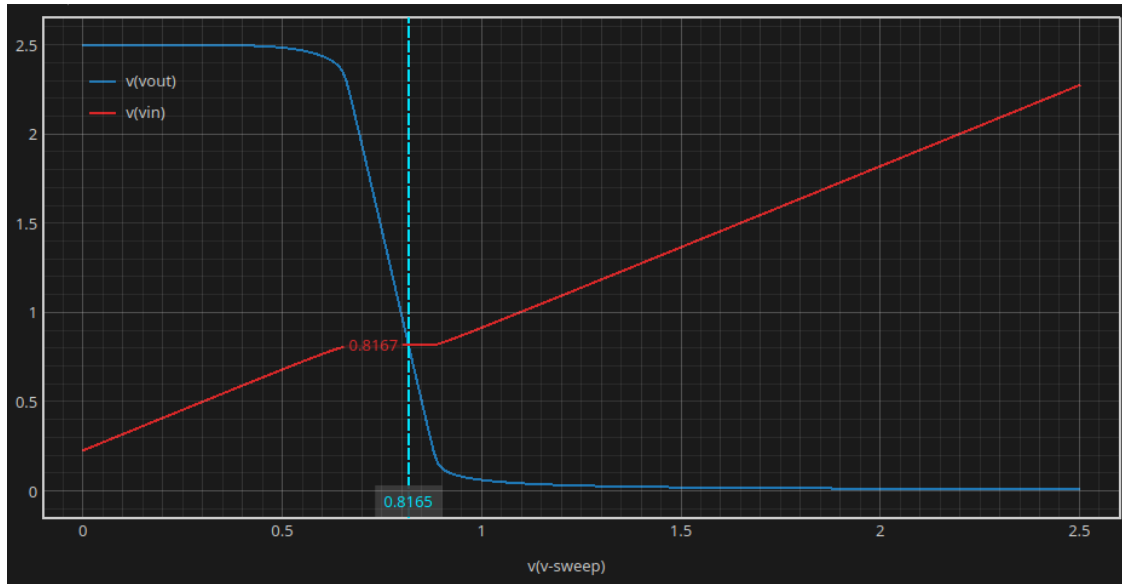


Figure 16: DC sweep of V_{OUT} and V_{IN} versus the applied signal V_{SIG} . The intersection occurs at exactly 0.8167 V.

Quiescent Operating Point

At what voltage do the V_{IN} and V_{OUT} curves cross?

As seen in Figure 16, the V_{OUT} and V_{IN} curves intersect at 0.8167 V. This specific voltage is the quiescent gate-source voltage (V_{GSQ}) of transistor M1. At this equilibrium point, no differential current flows through the feedback network (R_1 and R_2), forcing V_{OUT} to equal V_{IN} .

Is V_{OUT} vs. V_{SIG} linear?

We can divide the relationship between V_{out} and V_{SIG} into three distinct operating regions:

- **Cut-off Mode:** When $V_{SIG} < V_{TH}$, the MOSFET is in cut-off, resulting in $V_{out} = V_{DD} \approx 2.5$ V.
- **Saturation Region:** When $V_{SIG} \geq V_{TH}$ and $V_{GD} \leq V_{TH}$, the MOSFET operates in the saturation region. Because feedback improves linearity, this can be approximated as a linear region (characterized by a high slope and high gain). As current flows through the transistor, V_{out} decreases due to the voltage drop across the active load.
- **Triode Region:** When $V_{SIG} > V_{TH}$ and $V_{GD} > V_{TH}$, the MOSFET enters the triode region. The gain drops significantly, resulting in a very small slope and minimal variation of V_{out} with respect to V_{SIG} .

Consequently, the relationship between V_{out} and V_{SIG} is linear across the range of the amplifier.

3.3. Small-Signal Gain Linearity Validation

To determine if the gain is truly independent of the input amplitude, the derivative of the output voltage with respect to the input sweep (dV_{OUT}/dV_{SIG}) was plotted.

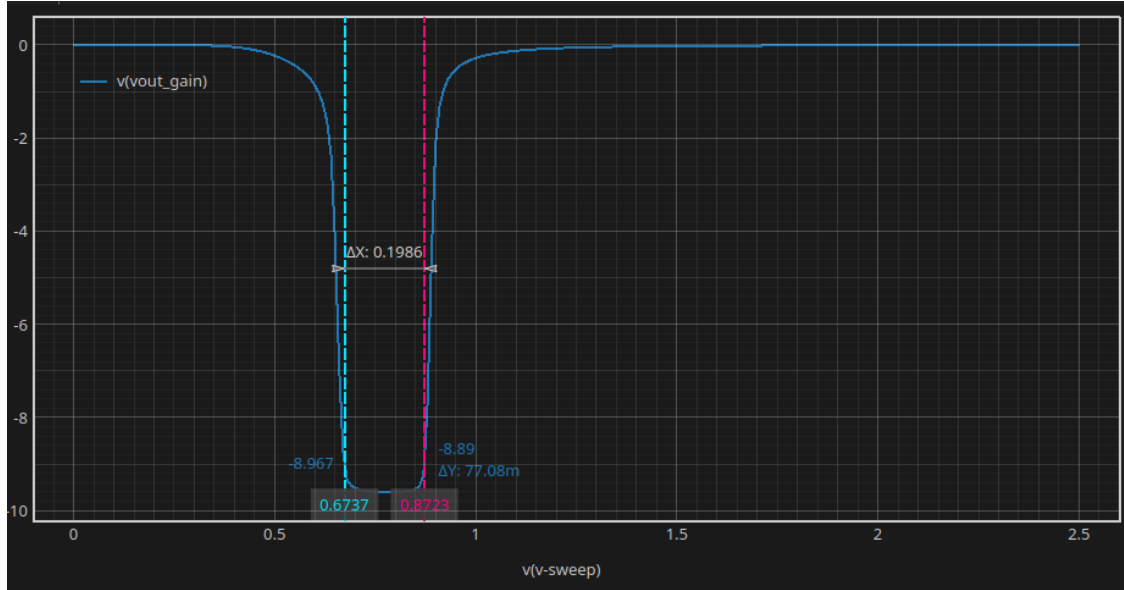


Figure 17: Derivative of V_{OUT} vs V_{SIG} . The flat plateau demonstrates a linear gain region of $\Delta X \approx 0.1986$ V, yielding a maximum stable gain of roughly -8.96 .

Analysis of Linear Region Limits

Is the gain linear in the operating range of the amplifier? Why?

The gain is constant in the range of amplifier because the negative feedback ensures that the closed-loop gain is dictated primarily by the ratio of the passive feedback resistors ($A_{CL} \approx -\frac{R_f}{R_{in}}$) rather than the highly variable, non-linear open-loop transconductance of the transistors.

Value of V_{in} and why?

This occurs because of a high open-loop gain and negative feedback makes a virtual ground at the input node. The feedback loop continuously adjusts the output voltage to minimize the error signal at the input terminal, effectively locking V_{IN} close to the quiescent V_{GS} required to sustain the intended bias current.

Analytical linear input range:

The amplifier fails when M1 or M2 exits saturation ($V_{DS} < V^*$). The output range before clipping is approximately $V_{DD} - 2V^* = 2.5 - 2(0.2) = 2.1$ V. The corresponding input range is:

$$\Delta V_{in} \approx \frac{V_{DD} - 2V^*}{|A_v|} = \frac{2.1}{10} = 210 \text{ mV} \quad (14)$$

For NMOS to be in sat:

$$V_{GS} \geq 668.4 \text{ mV} \quad \text{and} \quad V_{out} \geq V_{ov} = V_Q^* = 200 \text{ mV}$$

$$V_{out} \geq 0.2 \text{ V}$$

For PMOS to be in sat:

$$V_{SD} \geq V_{SG} - |V_{THP}|$$

$$V_{out} \leq V_G + |V_{THP}|$$

$$V_{out} \leq 1.5632 + 0.7749$$

$$V_{out} \leq 2.33 \text{ V}$$

So the range of V_{out} for the linear gain is $2.13 \text{ V} = 2.33 - 0.2$, then the range of the input that will be given by

$$\text{Range } V_{SIG} = \frac{\text{Range } V_{OUT}}{\text{Gain}} = \frac{2.13}{10} = 0.213 \text{ V}$$

Using rule,

$$\text{Range } V_{SIG} = \frac{V_{DD} - 2V_Q^*}{\text{Gain}} = \frac{2.5 - 2 \times 0.2}{10} = 0.21 \text{ V}$$

$$\text{Error}\% = \frac{0.21 - 0.213}{0.144} \times 100 = 1.43\%$$

3.4. Transient Transconductance (g_m) Stabilization

A 1 MHz AC sine wave with an appropriate DC offset= 0.8 was applied, and the transconductance of M1 was recorded over time.

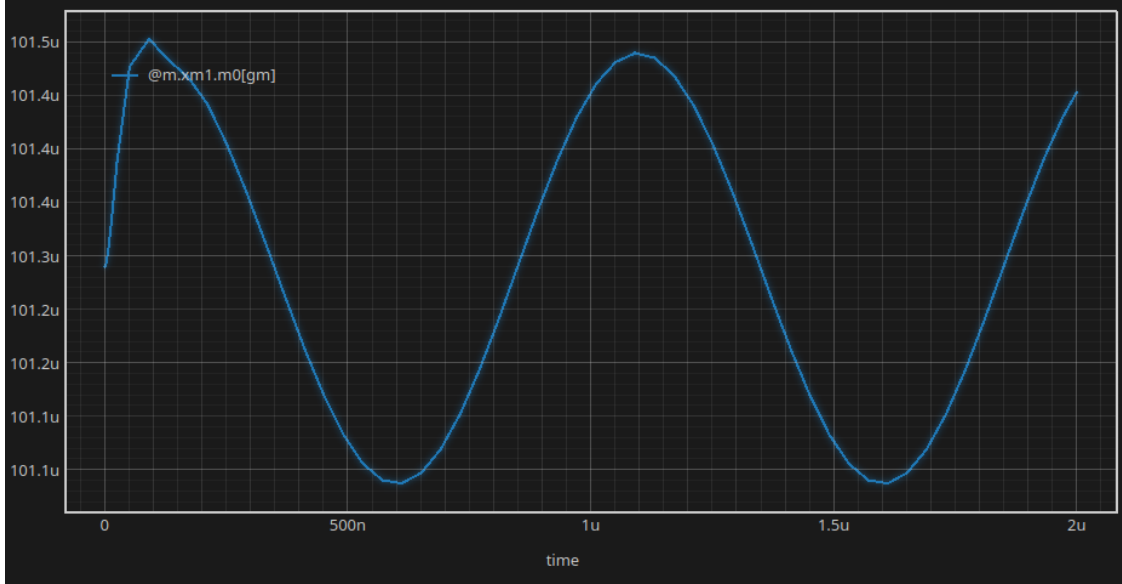


Figure 18: Transient transconductance (g_m) of M1 responding to a 1 MHz sine wave. The g_m fluctuates narrowly between 101.5 nA/V and 101.1 nA/V.

Impact of Feedback on Transconductance

In the open-loop configuration, large input swings actively modulate the device's overdrive voltage, causing g_m to vary dramatically and distorting the output signal. By employing negative feedback, the variation at the physical gate node (V_{IN}) is heavily suppressed. As shown in Figure 18, the peak-to-peak variation of g_m is restricted to a mere 0.4 $\mu\text{A}/\text{V}$ (approximately a 0.4% deviation). This steady g_m ensures a constant amplification factor across the entire signal period, effectively linearizing the analog amplifier.

4. [LAB 3] Part 3: Cascode for BW

4.1. Sizing (from Part 1)

Specifications. The common-source/cascode devices (M1, M2, M3) are sized using the Sizing Assistant (SA) to meet the Part 1 target specifications.

Table 2: Part 1 design specifications.

Parameter	Value
$A_v = g_m r_o$	50
g_m/I_D	10 S/A
Supply (V_{DD})	1.8 V
Quiescent output	$V_{DD}/2 = 0.9$ V
Bias current I_B	20 μ A

Length selection. With $I_D = 20 \mu\text{A}$, $g_m/I_D = 10 \text{ S/A}$, $V_{DS} = 0.9 \text{ V}$, $V_{SB} = 0$, sweeping L and plotting g_m/g_{ds} in the SA gives the curve below. Using the magic cursor, $L = 350 \text{ nm}$ is picked, giving $g_m/g_{ds} = 56.82$.

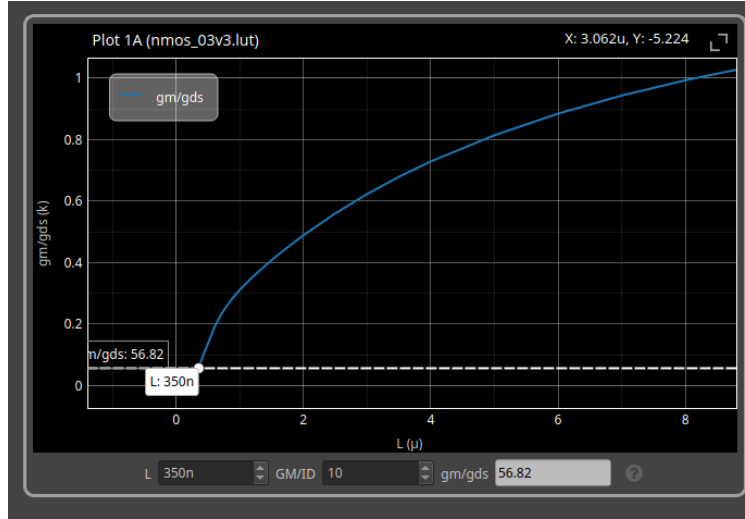


Figure 19: g_m/g_{ds} vs. L sweep in the Sizing Assistant; $L = 350 \text{ nm}$ picked ($g_m/g_{ds} = 56.82$).

Width selection. Feeding $L = 350 \text{ nm}$ back into the SA (with $I_D = 20 \mu\text{A}$, $g_m/I_D = 10 \text{ S/A}$, $V_{DS} = 0.9 \text{ V}$, $V_{SB} = 0$) gives:

Table 3: SA sizing result at $V_{DS} = 0.9 \text{ V}$, $V_{SB} = 0$.

Parameter	Value
W	3.48 μm (rounded to 3.5 μm)
L	350 nm
V_{GS}	806.4 mV
V_{DB}	900 mV

W	3.48u
VDB	900m
VGB	806.4m
VGS	806.4m
VDS	900m
VS	0
VGB	806.4m
VDB	900m

Figure 20: SA output giving $W = 3.48 \mu\text{m}$ for $L = 350 \text{ nm}$, $I_D = 20 \mu\text{A}$, $g_m/I_D = 10 \text{ S/A}$.

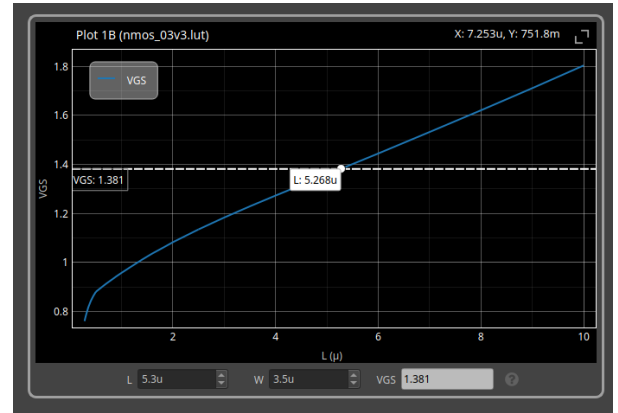
Magic-battery sizing (M4). V_B must equal $V_{GS3} + V_{DS2}$. Using the simulated cascode operating point (Section 4.3), $V_{GS3} = 921.6 \text{ mV}$ and $V_{DS2} = 460.2 \text{ mV}$, so the target is

$$V_B = V_{GS3} + V_{DS2} = 927 \text{ mV} + 450 \text{ mV} = 1.377 \text{ V} \quad (15)$$

M5 is diode-connected (same size as M1–M3: $W = 3.5 \mu\text{m}$, $L = 350 \text{ nm}$) and carries $I_B = 20 \mu\text{A}$, fixing its own $V_{GS} = 920.3 \text{ mV}$. M4 is kept in triode by sweeping its length L (with the same $W = 3.5 \mu\text{m}$) at $I_D = 20 \mu\text{A}$, $V_{DS} = V_{SB} = 0.45 \text{ V}$, and reading off the L that gives $V_{GS4} = V_B = 1.382 \text{ V}$.

	Name	TT-27.0
2	IG	N/A
3	L	350n
4	W	3.52u
5	VDB	900m
6	VGB	1.377
7	VGS	927m
8	VDS	450m
9	VS	450m

(a) $v_{gs} = 0.927$



(b) Refined sweep: $L = 5.3 \mu\text{m}$, $W = 3.5 \mu\text{m}$ gives $V_{GS} = 1.381 \text{ V} \approx V_B$

Figure 21: V_{GS} vs. L sweep for M4 (the “magic battery”), used to pick $L_4 = 5.3 \mu\text{m}$.

4.2. Circuit Construction

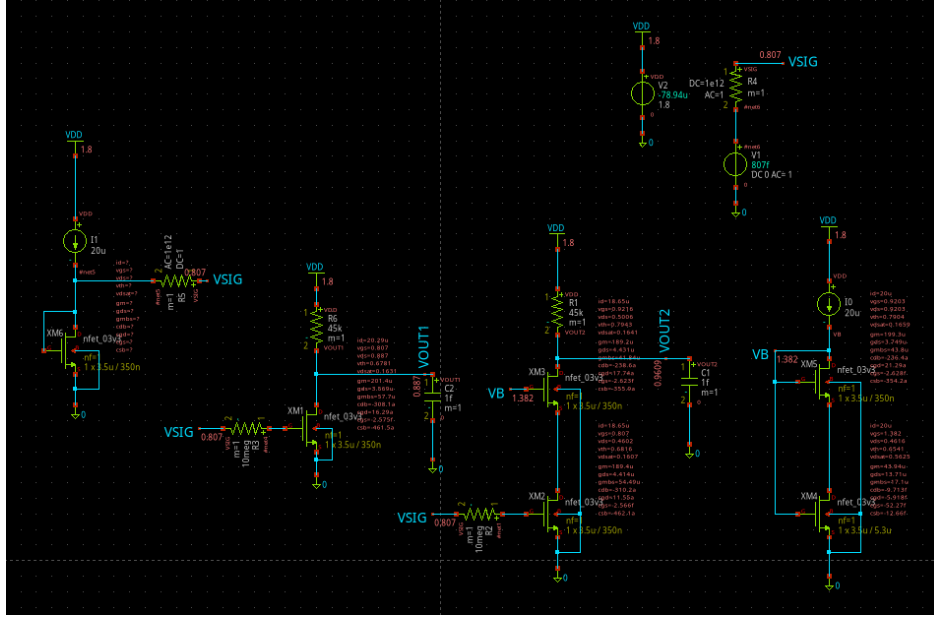


Figure 22: Schematic

Sizing R_D . With $I_{D,Q} \approx 20 \mu\text{A}$ and targeting a $V_{DD}/2$ drop across the resistor:

$$R_D = \frac{V_{DD}/2}{I_{D,Q}} = \frac{0.9 \text{ V}}{20 \mu\text{A}} = 45 \text{ k}\Omega \quad (16)$$

4.3. DC Operating Point Results

The DC OP simulation was run with $W = 3.5 \mu\text{m}$, $L = 350 \text{ nm}$ for M1–M3 (M5 the same), $R_D = 45 \text{ k}\Omega$, and $L_4 = 5.3 \mu\text{m}$ for the magic-battery device M4.

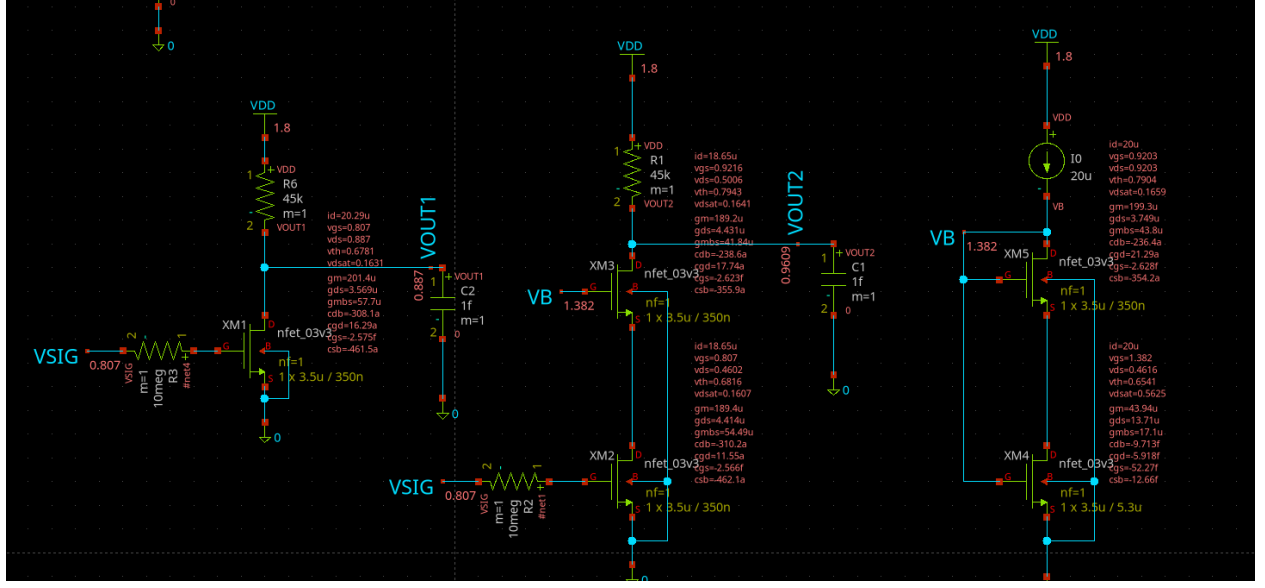


Figure 23: Annotated operating point of the CS amplifier (M1, left), the cascode amplifier (M2, M3, middle) and the magic-battery bias network (M4, M5, right).

Table 4: Combined DC operating-point parameters and parasitic capacitances for M1–M5 (resistive-load design).

Parameter	M1	M2	M3	M4	M5
W/L ($\mu\text{m}/\mu\text{m}$)	3.5/0.35	3.5/0.35	3.5/0.35	3.5/5.3	3.5/0.35
I_d (μA)	20.29	18.65	18.65	20.00	20.00
v_{gs} (mV)	887.0	807.0	921.6	1382	920.3
v_{ds} (mV)	887.0	460.2	500.6	461.6	920.3
v_{th} (mV)	678.1	681.6	794.3	654.1	790.4
V_{DSAT} (mV)	163.1	160.7	164.1	562.5	165.9
g_m ($\mu\text{A}/\text{V}$)	201.4	189.4	189.2	43.94	199.3
g_{ds} ($\mu\text{A}/\text{V}$)	3.569	4.414	4.431	13.71	3.749
g_{mb} ($\mu\text{A}/\text{V}$)	57.70	54.49	41.84	17.10	43.80
C_{gd}	16.29 aF	11.55 aF	17.74 aF	-5.918 fF	21.29 aF
C_{gs}	-2.575 fF	-2.566 fF	-2.623 fF	-52.27 fF	-2.628 fF
C_{db}	-308.1 aF	-310.2 aF	-238.6 aF	-9.713 fF	-236.4 aF
C_{sb}	-461.5 aF	-462.1 aF	-355.9 aF	-12.66 fF	-354.2 aF
C_{gg}	3.271 fF	3.264 fF	3.230 fF	61.11 fF	3.235 fF
C_{dd}	8.276 aF	3.761 aF	6.223 aF	6.870 fF	10.25 aF
C_{ss}	2.058 fF	2.049 fF	1.963 fF	41.04 fF	1.968 fF
Region	Sat (2)	Sat (2)	Sat (2)	Triode	Sat (2)

4.3.1 Design Validation

Table 5: Effective overdrive at the operating point.

	M1	M2	M3
$V_{ov} = V_{gs} - V_{th}$ (mV)	208.9	125.4	127.3

4.4. Saturation Check

Table 6: Saturation margin ($v_{ds} - V_{DSAT}$) – resistive-load design.

	M1	M2	M3
v_{ds} (mV)	887.0	460.2	500.6
V_{DSAT} (mV)	163.1	160.7	164.1
Margin (mV)	723.9	299.5	336.5

Q: Check that all transistors operate in saturation. Does any transistor operate in triode?

M1, M2, and M3 all report `region=2` (saturation) with $v_{ds} > V_{DSAT}$ and healthy positive margins. Only M4, in the magic-battery bias network, is intentionally kept in triode ($V_{DS4} = 461.6 \text{ mV} \ll V_{DSAT,4} = 562.5 \text{ mV}$) – its long channel length lets it act as a controllable, near-linear resistor so that a fixed L_4 generates exactly the V_{GS} needed to bias M3’s gate at V_B .

4.5. AC Analysis

4.5.1 Hand Analysis

The capacitor values are from ADT.

CS Amplifier (M1).

$$r_{o1} = \frac{1}{g_{ds1}} = \frac{1}{3.569 \mu\text{A/V}} = 280.2 \text{ k}\Omega, \quad R_{out1} = r_{o1} \parallel R_D = 38.77 \text{ k}\Omega \quad (17)$$

$$A_{v1} = -g_m[1]R_{out1} = -201.4 \mu\text{A/V} \times 38.77 \text{ k}\Omega = -7.807 \text{ (17.85 dB)} \quad (18)$$

The parameter C_{gg} represents the total gate capacitance. Based on the extracted parameters in `image05801c.png`, we can directly use the exact C_{gd} and C_{gg} values instead of estimating them:

$$C_{gg1} = 4.524 \text{ fF}, \quad C_{gd1} = 601.4 \text{ aF} = 0.6014 \text{ fF} \quad (19)$$

This C_{gd1} is Miller-multiplied and, together with the $10 \text{ M}\Omega$ source resistance, sets the dominant input pole:

$$C_{in,1} = C_{gg1} + C_{gd1}|A_{v1}| = 4.524 \text{ fF} + 0.6014 \text{ fF}(7.807) = 9.219 \text{ fF} \quad (20)$$

$$BW_1 = \frac{1}{2\pi R_{sig} C_{in,1}} = \frac{1}{2\pi(10 \text{ M}\Omega)(9.219 \text{ fF})} = 1.726 \text{ MHz} \quad (21)$$

$$GBW_1 = |A_{v1}| \times BW_1 = 7.807 \times 1.726 \text{ MHz} = 13.47 \text{ MHz} \quad (22)$$

Cascode Amplifier (M2, M3).

$$r_{o2} = \frac{1}{g_{ds2}} = 226.6 \text{ k}\Omega, \quad r_{o3} = \frac{1}{g_{ds3}} = 225.7 \text{ k}\Omega \quad (23)$$

Boosted output resistance (paralleled with R_D , which dominates):

$$R_{out,intrinsic} = r_{o2} + r_{o3} + (g_m[3] + g_{mb3})r_{o2}r_{o3} = 12.27 \text{ M}\Omega, \quad R_{out2} = R_{out,intrinsic} \parallel R_D = 44.84 \text{ k}\Omega \quad (24)$$

$$A_{v2} = -g_m[2]R_{out2} = -189.4 \mu\text{A/V} \times 44.84 \text{ k}\Omega = -8.492 \text{ (18.58 dB)} \quad (25)$$

Applying the exact parameters from `image05801c.png` to M2 : $C_{gg2} = 4.524 \text{ fF}$, $C_{gd2} = 0.6014 \text{ fF}$ (26) The local gain from M2's gate to node V_X (M2's drain) is small since V_X is loaded by M3's low source-resistance. This suppresses the Miller multiplication of C_{gd} :

$$A_{v,local} = -g_m[2](r_{o2} \parallel R_{in3}) = -189.4 \mu\text{A/V} \times \left(226.6 \text{ k}\Omega \parallel \frac{1}{189.2 \mu\text{A/V} + 41.84 \mu\text{A/V}} \right) = -0.804 \quad (27)$$

$$C_{in,2} = C_{gg2} + C_{gd2}|A_{v,local}| = 4.524 \text{ fF} + 0.6014 \text{ fF}(0.804) = 5.008 \text{ fF} \quad (28)$$

$$BW_2 = \frac{1}{2\pi R_{sig} C_{in,2}} = \frac{1}{2\pi(10 \text{ M}\Omega)(5.008 \text{ fF})} = 3.178 \text{ MHz} \quad (29)$$

$$GBW_2 = |A_{v2}| \times BW_2 = 8.492 \times 3.178 \text{ MHz} = 26.99 \text{ MHz} \quad (30)$$

4.5.2 Simulation Results

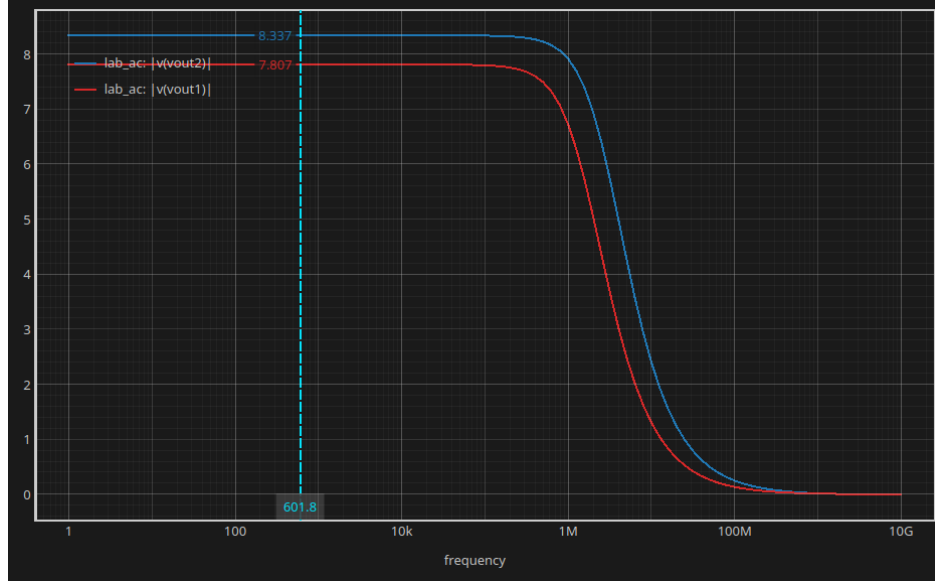


Figure 24: Magnitude response: CS amplifier (V_{OUT1} , red) and cascode amplifier (V_{OUT2} , blue).

```
Using SPARSE 1.3 as Direct Linear Solver
Note: Starting dynamic gmin stepping
Note: Dynamic gmin stepping completed

No. of Data Rows : 101
dc_gain_1      = 7.806966e+00
bw_1           = 1.694538e+06
ugf_1          = 1.318210e+07
dc_gain_2      = 8.336792e+00
bw_2           = 3.007130e+06
ugf_2          = 2.487425e+07
binary raw file "lab_3.raw"
ngspice 1 -> █
```

Figure 25: ngspice .meas output: DC gain, BW, and UGF for both stages.

Table 7: AC simulation results (measured via .meas).

Quantity	CS (Stage 1)	Cascode (Stage 2)
DC Gain (V/V)	7.807	8.336
DC Gain (dB)	17.85	18.42
BW	1.6945 MHz	3.0071 MHz
UGF	13.182 MHz	24.874 MHz
GBW ($A_v \times BW$)	13.229 MHz	25.067 MHz

4.5.3 Comparison: Simulation vs. Hand Analysis

Table 8: CS amplifier (M1), resistive load: simulation vs. hand analysis.

Quantity	Hand Analysis	Simulation	Error (%)
DC Gain (V/V)	7.807	7.807	0.00
DC Gain (dB)	17.85	17.85	0.00
BW (MHz)	1.726	1.695	1.83
GBW (MHz)	13.47	13.23	1.81

Table 9: Cascode amplifier (M2, M3), resistive load: simulation vs. hand analysis.

Quantity	Hand Analysis	Simulation	Error (%)
DC Gain (V/V)	8.492	8.336	1.87
DC Gain (dB)	18.58	18.42	0.87
BW (MHz)	3.178	3.007	5.69
GBW (MHz)	26.99	25.07	7.67

Q: Comment on the results.

The DC-gain hand analysis, based purely on g_m and g_{ds} from the OP simulation, matches the simulated gain almost exactly for the CS stage and within 2% for the cascode stage – confirming the resistive-load design point.

The AC simulation clearly demonstrates the secondary major advantage of the cascode topology: **bandwidth extension via reduced Miller effect**. Driven by a high-impedance source ($R_{sig} = 10\text{ M}\Omega$), the plain CS amplifier’s bandwidth is limited to 1.69 MHz because C_{gd1} is multiplied by the full stage gain (≈ 7.8), creating a large effective input capacitance. The cascode amplifier, despite delivering essentially the same (slightly higher) DC gain, achieves nearly **1.8× the bandwidth** (3.01 MHz) because the cascode device M3 clamps node V_X to a low impedance, so the local gain seen by C_{gd2} is small and Miller multiplication is suppressed. As a result the GBW nearly doubles, from 13.2 MHz to 25.1 MHz, and the UGF closely tracks the GBW in both cases, confirming a single dominant (input) pole in each amplifier.